

OpenDevin AI Platform- DevOps Local Deployment Documentation

Objective

■ Tools & Technologies Used

▪ Category	Tool / Version
▪ Operating System	Ubuntu 22.04 (WSL 2 on Windows 10)
▪ Containerization	Docker 24+, Docker Compose v2.20
▪ Configuration Management	.env files + Docker Compose
▪ Automation	Shell script (dev.sh) + Poetry
▪ Languages	Python 3.12, Node.js 18
▪ Dependency Managers	Poetry (Python), npm (Node.js)
▪ Version Control	Git

Environment Where It Was Built

- Platform: Windows 10 Pro
- Virtualization: WSL 2 (Ubuntu 22.04)
- Editor: Visual Studio Code
- Terminal: Windows Terminal (WSL mode)

DevOps Steps Followed

1.System Preparation

```
> sudo apt update && sudo apt install -y git curl docker.io docker-compose  
python3.12 python3.12-venv python3.12-dev
```

```
> sudo usermod -aG docker $USER && newgrp docker
```

2. Install Node.js via nvm

```
> curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.3/install.sh |  
bash
```

```
> source ~/.bashrc.
```

```
> nvm install 18
```

```
> nvm use 18
```

3. Install Poetry

```
> curl -sSL https://install.python-poetry.org | python3 -  
source $HOME/.local/bin/poetry
```

4. Clone the Repository

```
> git clone https://github.com/OpenDevin/OpenDevin.git  
cd OpenDevin
```

5. Set Up Configuration

```
> cp .env.example .env
```

```
> nano .env
```

```
> Sample .env values:
```

```
OPENAI_API_KEY=sk-xxx
```

```
DEVIN_MODE=local
```

```
REDIS_URL=redis://localhost:6379
```

6. Install Dependencies

```
> poetry install  
npm install
```

7. Docker Containerization

```
> docker-compose.yml created to define app and redis services.  
> Run: docker-compose up --build
```

8. Automation Script (dev.sh)

```
> #!/bin/bash  
  
poetry install  
npm install  
docker-compose up --build
```

```
chmod +x dev.sh  
./dev.sh
```

9. Access the Platform

```
> Open browser and navigate to http://localhost:3000
```

Testing the Application

```
> poetry run pytest
```

■ Troubleshooting Table

Issue	Solution
▪ Docker permission denied	<code>sudo usermod -aG docker \$USER &&</code> <code>USER && newgrp docker</code>
▪ Poetry not found	<code>source \$HOME/.local/bin/poetry</code>
▪ API Key invalid	Check .env file and restart container

DevOps Practices Summary

Practice	How It Was Implemented
▪ Automation	Setup using dev.sh script
▪ Containerization	Docker Compose for multi-container apps
▪ Config Management	.env files + Docker env_file usage
▪ Documentation	This Word document

Final Notes

Result: Fully working local setup of OpenDevin AI Agent

- Submission Purpose: DevOps Job Assessment
- Built by: Nitish Sharma

Contacts

Name: Nitish Sharma

GitHub: <https://github.com/nitish-sharma>

LinkedIn: <https://linkedin.com/in/nitishsharma>